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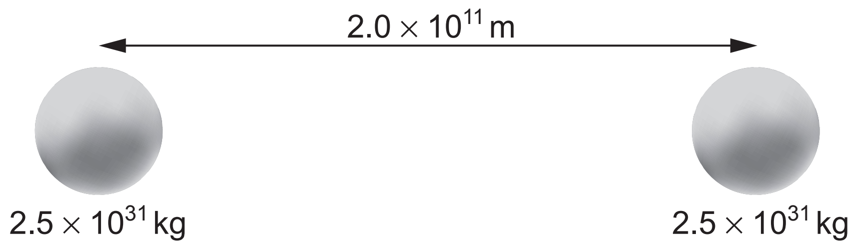
(b) A two-star system has two stars of equal mass as shown.



(i) State or calculate the position of the centre of mass of the two-star system. [1]

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(ii) Calculate the period of orbit of the stars about their centre of mass. [2]

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(iii) Calculate the maximum red shift (or blue shift) of the hydrogen 434 nm line due to the orbital motion of the stars. [3]

Not drawn to scale

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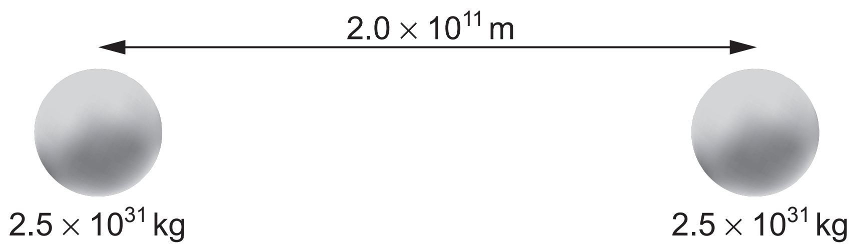
(c) The centre of mass of this two-star system is also the point where the resultant gravitational field strength is zero. Explain why these points only coincide when the stars have identical masses. [2]



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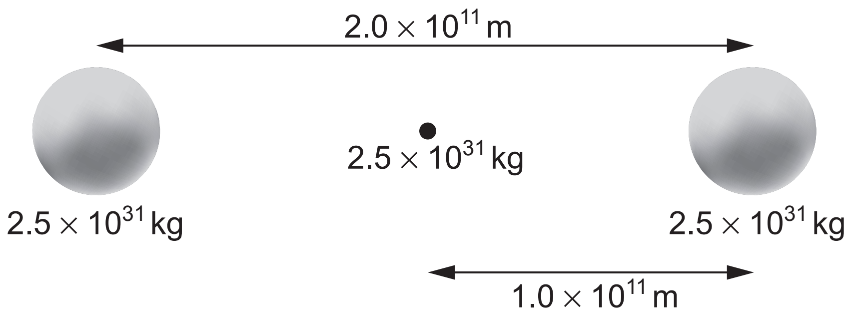
(d) A different star system consists of three stars all of equal mass. Two stars orbit around a stationary black hole and the black hole is always halfway between the two stars as shown.



(i) Explain why the resultant force on the black hole is always zero. [1]

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(ii) Explain why the gravitational force acting on either of the orbiting stars is five times greater in this three-star system than the two-star system of part (b). [2]

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(iii) Joseff claims that the stars in the three-star system will provide a red shift that is five times larger than that of the two-star system of part (b). Evaluate whether or not he is correct. [3]

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(e) A recent theoretical publication suggests that the decay of the Higgs Boson will give direct evidence for dark matter. Suggest what needs to be done for this theory to be generally accepted by scientists in the future. [2]

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